# Analyze and select, based on scientific and technical literature published in recent years, ten (10) years, machine learning and computer vision models oriented to the detection of diseases and pests in rice crops, prioritizing criteria of computational efficiency, operation in offline mode and compatibility with entry-level mobile devices. a. Deliverable: i. Benchmarking executive report, including comparison of Models, advantages, limitations and technical justification of the proposals selected.

Recent scientific literature highlights lightweight deep learning models like MobileNet variants and YOLO adaptations as top performers for rice disease and pest detection on resource-constrained devices. These prioritize low parameters, FLOPs, offline inference via ONNX/TensorFlow Lite, and mobile compatibility while achieving high accuracy on datasets like Rice Leaf Diseases.[1][2][3]

## Model Selection Criteria

Models were selected from 2016-2026 publications emphasizing CNNs and object detectors for rice-specific tasks. Key metrics include parameters (<10M preferred), FLOPs (<1G for mobiles), accuracy/mAP >90%, and explicit mobile/offline mentions. Excluded heavy models like standard ResNet50 despite good accuracy (92.5%) due to high compute.[4][5][6]

## Benchmark Comparison

| Model | Base | Accuracy/mAP | Params | FLOPs | Mobile | Offline |
|---|---|---|---|---|---|---|

| | Architecture | | (M) | (G) | FPS | Support |
|---|---|---|---|---|---|---|
| LiSA-MobileNetV2 [7] | MobileNetV2 + Attention | 95.7% | ~2.0 | ~0.4 | 30+ | Yes (TFLite) |
| MobileNet-CA-YOLO [8] | YOLOv7 + MobileNetV3 | 93.7% mAP | ~5-7 | ~1.0 | High | Yes |
| TLI-YOLO (YOLOv8n) [9] | YOLOv8 + Transfer | ~95% | 3.2 | Low | 30 | Yes (Android) |
| GE-ShuffleNet [6] | ShuffleNetV2 + Ghost | 96.6% | 5.9 | Low | High | Yes |
| SSD-YOLO [10] | YOLO + SSD lightweight | 98.99% (blast) | 6.0 | Low | High | Yes |
| EfficientNet-B0 [11] | EfficientNet | 96%+ | 5.3 | 0.39 | 25+ | Yes (TFLite) |
| MobileNetV3 Large [4] | MobileNetV3 | 96.3% | ~12 | ~0.3 | High | Yes |
| Lightweight CNN (2018) [3] | Custom two-stage | High (mobile-tuned) | Small | Low | Mobile-suited | Yes |
| Improved ShuffleNet V2 [6] | ShuffleNet | 94.4% | Low | Low | High | Yes |
| Rice-DP-YOLO (YOLOv8n) [12] | YOLOv8n improved | 77.8% (15 classes) | ~3 | Low | High | Yes |

Data aggregated from benchmarks; FPS on entry-level Android (e.g., Snapdragon 6xx).[10][8][9][6][7][12][4]

## Advantages and Limitations

**Advantages:** These models excel in edge deployment with depthwise convolutions (MobileNet/ShuffleNet) reducing compute by 50-75% vs. VGG/ResNet, enabling 20-30 FPS offline on mobiles without internet. YOLO variants add real-time pest bounding boxes for multi-class detection. Attention (CA/SE) boosts accuracy 2-6% on small datasets.[3][8][7]

**Limitations:** Smaller models trade ~1-5% accuracy for speed; struggle with tiny pests or diverse lighting (mAP drops 5-10%). Limited multi-pest coverage in some (focus on 3-6 diseases); need fine-tuning for local rice varieties.[9][13][10]

## Technical Justification

Selections favor inverted residuals (MobileNetV3/EfficientNet) and ghost modules (GE-ShuffleNet) for parameter efficiency, proven in rice datasets (e.g., 3000+ images). YOLO adaptations suit pests needing localization, with SIoU loss improving small-object mAP. All convert to TFLite for offline mobile use, aligning with entry-level hardware (2-4GB RAM). Top picks: LiSA-MobileNetV2 for classification, MobileNet-CA-YOLO for detection.[2][11][8][7]

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1. https://pmc.ncbi.nlm.nih.gov/articles/PMC10088765/
2. https://pmc.ncbi.nlm.nih.gov/articles/PMC11253532/
3. https://ar5iv.labs.arxiv.org/html/1812.01043
4. https://d-nb.info/1371393532/34
5. https://arxiv.org/html/2501.06740v1
6. https://www.techscience.com/cmc/v75n2/52135
7. https://pmc.ncbi.nlm.nih.gov/articles/PMC12370686/
8. https://ui.adsabs.harvard.edu/abs/2023Agric..13.1285J/abstract
9. https://pmc.ncbi.nlm.nih.gov/articles/PMC12031063/
10. https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2025.1643096/full
11. https://www.arxiv.org/pdf/2408.01752.pdf
12. https://pmc.ncbi.nlm.nih.gov/articles/PMC12528418/
13. https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2024.1416940/full
14. https://www.atlantis-press.com/article/126004097.pdf
15. https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0313387
16. https://arccjournals.com/journal/indian-journal-of-agricultural-research/A-6061
17. https://www.rcs.cic.ipn.mx/2023_152_10/Pests Detection in Agricultural Crops Using Computer Vision.pdf

18. https://www.ijsat.org/papers/2025/2/4410.pdf

19. https://repositorio.iscte-iul.pt/bitstream/10071/26147/1/article_90301.pdf

20. https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2025.1619365/full